Shawn Xiao

Email: s7xiao@ucsd.edu

PID: A18276668

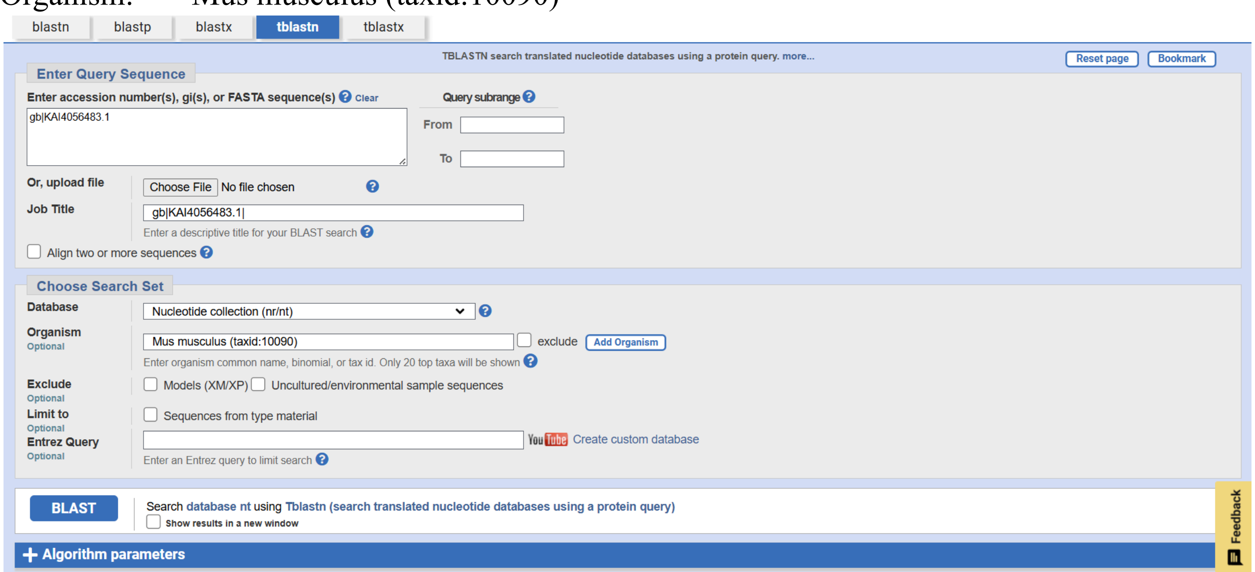
1. Name: PIEZO 1

   Accession: KAI4056483.**1**

   Species: Homo Sapiens

   Known function: PIEZO 1 protein is a mechanosensitive ion channel protein that allows calcium ions to transfer for muscle contraction in hearts.

2. Method: TBLASTN search against nt
   Database: Nucleotide collection (nt)
   Organism: Mus musculus (taxid:10090)



‹ Edit Search | Save Search | Search Summary ⌄ | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Your search is limited to records **include: Mus musculus (taxid:10090)**

Job Title	gb|KAI4056483.1|
RID	FX7F5V68014 *Search expires on 10-28 06:17 am* Download All ⌄
Program	TBLASTN Citation ⌄
Database	nt See details ⌄
Query ID	KAI4056483.1
Description	piezo type mechanosensitive ion channel component 1 [H...
Molecule type	amino acid
Query Length	2521
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download ⌄ Select columns ⌄ Show 100

select all *80 sequences selected* GenBank Graphics

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1), transcript variant 2, mRNA	Mus musculus	3488	3488	100%	0.0	70.48%	8216	NM_001037298.1
Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1), transcript variant 1, mRNA	Mus musculus	3487	3487	100%	0.0	70.41%	8219	NM_001357349.1

Chosen match: Accession NM_001037298.1, a 8216 base pair Mus musculus mRNA. See below for more alignment details.

This alignment has an e-value of 0.0, 100% coverage, and 70.48% identity.

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Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1), transcript variant 2, mRNA

Sequence ID: NM_001037298.1 Length: **8216** Number of Matches: **1**

Range 1: 292 to 7929 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
3488 bits(9044)	0.0	Compositional matrix adjust.	2054/2564(80%)	2215/2564(86%)	61/2564(2%)	+1

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Query  1     MEPHVLGAVLYWlllpcallaacllRFSGlslvyllflllPWFPGPTRCGLQGHTGRll  60
             MEPHVLGA LYWLLLPC LLAA LLRF+ LSLVYLLFLLLLPW PGP+R  + GHTGRLL
Sbjct  292   MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL  471

Query  61    rallglsllflvahlalQICLHTVPRLDQLLGPSCSRWETLSRHIGVTRLDLKDIPNAIR  120
             RALL LSLLFLVAHLA QICLHTVP LDQ LG + S W  +S+HIGVTRLDLKDI N  R
Sbjct  472   RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  651

Query  121   LVAPDLGILVVSSVCLGICGRLARNTRQSPHPRELDDDERDVD------ASPTAGLQEAA  174
             LVAPDLG+L+ SS+CLG+CGRL R  RQS   +ELDDD+ D D      A+P  GL+ A
Sbjct  652   LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQELDDDDDDDDDEDIDAAPAVGLKGAP  831

Query  175   TLAPTRRSRLAARFRVTAHWLLVAAGRVLAVTLLALAGIAHPsalssvylllflalCTWW  234
              LA  RR  LA+RFRVTAHWLL+ +GR L + LLALAGIAHPSA SSVYL++FLA+CTWW
Sbjct  832   ALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTWW  1011

Query  235   ACHFPISTRGFSRLCAAVGCFGAGHLICLYCYQMPLAQALLPPAGIWARVLGLKDFVGPT  294
             +CHFP+S  GF+ LC  V CFGAGHLICLYCYQ P  Q +LPP  IWAR+ GLK+FV
Sbjct  1012  SCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDLP  1191

Query  295   NCSSPHALVLNTGLDWPVYASPGVLLLLCYATASLRKLRAYRPSGQRKEAAKGYEARELE  354
             N SSP+ALVLNT   WP+Y SPG+LLLL Y   SL KL    PS  RKE  +  E  ELE
Sbjct  1192  NYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHELE  1371

Query  355   LAELDQWPQERESDQHVVPTAPDTEADNCIVHELTGQSSLLRRPVRPKRAEPGEASPLHS  414
             L  L+  PQ R++ Q  +P   + + DNC VH LT QS + +RPVRP+ AE  E SPLH
Sbjct  1372  LDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLHG  1551

Query  415   LGHLIMDQSYVCALIAMMVWSITYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLYG  474
             LGHLIMDQSYVCALIAMMVWSI YHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLYG
Sbjct  1552  LGHLIMDQSYVCALIAMMVWSIMYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLYG  1731

Query  475   MTLCCLRYVWAMDLRPELPTTLGPVSLRQLGLEHTRYPCLDLGAMllytltfwlllRQFV  534
             +TLCCLRYVWAM+L PELPTTLGPVSL QLGLEHTRYPCLDLGAMLLY LTFWLLLRQFV
Sbjct  1732  LTLCCLRYVWAMEL-PELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFV  1908

Query  535   KEKLLKWAESPAALTEVTVADTEPTRTQTLLQSLGELVKGVYAKYWIYVCAGMFIVVSFA  594
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Sbjct  1909  KEKLLKKQKVPAALLEVTVADTEPTQTQTLLRSLGELVTGIYVKYWIYVCAGMFIVVSFA  2088
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Related Information

Gene - associated gene details

GEO Profiles - microarray expression data

Genome Data Viewer - aligned genomic context

>Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1), transcript variant 2, mRNA
Sequence ID: NM_001037298.1 Length: 8216

```
Range 1: 292 to 7929

Score:3488 bits(9044), Expect:0.0,
Method:Compositional matrix adjust.,
Identities:2054/2564(80%), Positives:2215/2564(86%), Gaps:61/2564(2%)

Query  1     MEPHVLGAVLYWlllpcallaacllRFSGlslvyllflllllPWFPGPTRCGLQGHTGRll  60
             MEPHVLGA LYWLLLPC LLAA LLRF+ LSLVYLLFLLLLPW PGP+R  + GHTGRLL
Sbjct  292   MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL  471

Query  61    rallglsllflvahlalQICLHTVPRLDQLLGPSCSRWETLSRHIGVTRLDLKDIPNAIR  120
             RALL LSLLFLVAHLA QICLHTVP LDQ LG + S W  +S+HIGVTRLDLKDI N  R
Sbjct  472   RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  651

Query  121   LVAPDLGILVVSSVCLGICGRLARNTRQSPHPRELDDDERDVD------ASPTAGLQEAA  174
             LVAPDLG+L+ SS+CLG+CGRL R  RQS   +ELDDD+ D D      A+P  GL+ A
Sbjct  652   LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQELDDDDDDDDDEDIDAAPAVGLKGAP  831

Query  175   TLAPTRRSRLAARFRVTAHWLLVAAGRVLAVTLLALAGIAHPsalssvyllllflalCTWW  234
              LA  RR  LA+RFRVTAHWLL+ +GR L + LLALAGIAHPSA SSVYL++FLA+CTWW
Sbjct  832   ALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTWW  1011

Query  235   ACHFPISTRGFSRLCAAVGCFGAGHLICLYCYQMPLAQALLPPAGIWARVLGLKDFVGPT  294
             +CHFP+S  GF+ LC  V CFGAGHLICLYCYQ P  Q +LPP  IWAR+ GLK+FV
Sbjct  1012  SCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDLP  1191

Query  295   NCSSPHALVLNTGLDWPVYASPGVLLLLCYATASLRKLRAYRPSGQRKEAAKGYEARELE  354
             N SSP+ALVLNT   WP+Y SPG+LLLL Y   SL KL    PS  RKE  +  E  ELE
Sbjct  1192  NYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHELE  1371

Query  355   LAELDQWPQERESDQHVVPTAPDTEADNCIVHELTGQSSLLRRPVRPKRAEPGEASPLHS  414
             L  L+  PQ R++ Q  +P   + + DNC VH LT QS + +RPVRP+ AE  E SPLH
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Query  535   KEKLLKWAESPAALTEVTVADTEPTRTQTLLQSLGELVKGVYAKYWIYVCAGMFIVVSFA  594
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Query  655   YWRNLTGFTDEQLGDLGLEQFSVSELFSSILVPGFFLLACILQLHYFHRPFMQLTDMEHV  714
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Sbjct  2269  YWRNLTGFTDEQLGDLGLEQFSVSELFSSILIPGFFLLACILQLHYFHRPFMQLTDLEHV  2448

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Query  775   PEG-AAKWGLVAERLLELAAGFSDVLSRVQVFLRRLLELHVFKLVALYTVWVALKEVSVM  833
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Sbjct  2593  PEGTASKWGLVADRLLDLAASFSAVLTRIQVFVRRLLELHVFKLVALYTVWVALKEVSVM  2772

Query  834   NLLLVVLWAFALPYPRFRPMASCLSTVWTCVIIVCKMLYQLKVVNPQEYSSNCTEPFPNS  893
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Sbjct  2773  NLLLVVLWAFALPYPRFRPMASCLSTVWTCIIIVCKMLYQLKIVNPHEYSSNCTEPFPNN  2952

Query  894   TNLLPTEISQSLLYRGPVDPANWFGVRKGFPNLGYIQNHLQVLLLLVFEAIVyrrqehyr  953
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Query  1014  LVTLHGCWLVAILTRRHRQAIARLWPNyclflalfllyqyllclGMPPALCIDYPWRWSR  1073
             +V LHGCWLVAILTRR R+AIARLWPNYCLFL LFLLYQYLLCLGMPPALCIDYPWRWS+
Sbjct  3313  MVILHGCWLVAILTRRRREAIARLWPNYCLFLTLFLLYQYLLCLGMPPALCIDYPWRWSK  3492

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Sbjct  3493  AIPMNSALIKWLYLPDFFRAPNSTNLISDFLLLLCASQQWQVFSAERTEEWQRMAGINTD  3672

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              LEPLRGEPNP+PNFIHCRSYLDMLKVAVFRYLFWLVLVVVFV GATRISIFGLGYLLAC
Sbjct  3673  HLEPLRGEPNPIPNFIHCRSYLDMLKVAVFRYLFWLVLVVVFVAGATRISIFGLGYLLAC  3852

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             FYLLLFGT LLQ+DTRA+LVLWDCLILYNVTVIISKNMLSLL+CVFVEQMQ+ FCWVIQL
Sbjct  3853  FYLLLFGTTLLQKDTRAQLVLWDCLILYNVTVIISKNMLSLLSCVFVEQMQSNFCWVIQL  4032

Query  1254  FSLVCTVKGYYDPKEMMDRDQDCLLPVEEAGIIWDSVCffflllqrrvflSHYYLHVRAD  1313
             FSLVCTVKGYYDPKEMM RD+DCLLPVEEAGIIWDS+CFFFLLLQRR+FLSHY+LHV AD
Sbjct  4033  FSLVCTVKGYYDPKEMMTRDRDCLLPVEEAGIIWDSICFFFLLLQRRIFLSHYFLHVSAD  4212

Query  1314  LQATALLASRGFALYNAANLKSIDFHRRIEEKSLAQLKRQMERIRAKQEKHRQGRVDRSR  1373
             L+ATAL ASRGFALYNAANLKSI+FHR+IEEKSLAQLKRQM+RIRAKQEK+RQ +  R +
Sbjct  4213  LKATALQASRGFALYNAANLKSINFHRQIEEKSLAQLKRQMKRIRAKQEKYRQSQASRGQ  4392

Query  1374  PQDTLGPKDpglepgpdspggssppRRQWWRPWLDHATVIHSGDYFLFesdseeeeeavp  1433
              Q    P+DP  EPGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEEEEA+P
Sbjct  4393  LQSK-DPQDPSQEPGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEEEEALP  4569

Query  1434  edpRPSAQSAFQLAYQAWVTNaqavlrrrqqeqeqarqeqagqLPTGGGPSQEVepaegp  1493
             EDPRP+AQSAFQ+AYQAWVTN        +Q +E+ARQE+A QL +GG  + +VEP + P
Sbjct  4570  EDPRPAAQSAFQMAYQAWVTN---AQTVLRQRRERARQERAEQLASGGDLNPDVEPVDVP  4740

Query  1494  eeaaagRSHVVQRVLSTAQFLWMLGQALVDELTRWLQEFTRHHGTMSDVLRAERYLLTQE  1553
             E+  AGRSH++QRVLST QFLW+LGQA VD LTRWL+ FT+HH TMSDVL AERYLLTQE
Sbjct  4741  EDEMAGRSHMMQRVLSTMQFLWVLGQATVDGLTRWLRAFTKHHRTMSDVLCAERYLLTQE  4920

Query  1554  LLQGGEVHRGVLDQLYTSQAEATLPGPTEAPNAPSTVSSGLGAEEPLSSMTDDMGSPLST  1613
             LL+ GEV RGVLDQLY  + EATL GP E  + PST SSGLGAEEPLSSMTDD  SPLST
Sbjct  4921  LLRVGEVRRGVLDQLYVGEDEATLSGPMETRDGPSTASSGLGAEEPLSSMTDDTSSPLST  5100
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             GY+TRSGSEE VTD G+ +AG SL+         +  MRTASELLLDRRL IPELEEAE
Sbjct  5101  GYNTRSGSEEIVTDAGDLQAGTSLHGSQELLANARTRMRTASELLLDRRLHIPELEEAER  5280

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Query  1725  PSKRFWMTAIVFTEIAVVVKYLFQFGFFPWNSHVVLRRYENKPYFPPRILGLEKTDGYIK  1784
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Query  1785  YDLVQLMALFFHRSQLLCYGLWDHEEDSPSKEHDKSgeeeqgaee---------------  1829
             YDLVQLMALFFHRSQLLCYGLWDHEED   K+H +S  +++ A+E
Sbjct  5641  YDLVQLMALFFHRSQLLCYGLWDHEEDRYPKDHCRSSVKDREAKEEPEAKLESQSETGTG  5820

Query  1830  --gpgVPAATTEDHIQVEARVGPTDGTPEPQVELRPrdtrrislrfrrrKKEGPARKGaa  1887
                  V A T  DHIQ +  +   D   +P  +L+P    R      RR+KE P  KG A
Sbjct  5821  HPKEPVLAGTPRDHIQGKGSIRSKDVIQDPPEDLKP-RHTRHISIRFRRRKETPGPKGTA  5997

Query  1888  aieaedreeeegeeekeaPTGREKrpsrsggrvraagrrLQGFCLSLAQGTYRPLRRFFH  1947
              +E E  E E  E  +       +  S+   R++AAGRRLQ FC+SLAQ  Y+PL+RFFH
Sbjct  5998  VMETEHEEGEGKETTERKRPRHTQEKSKFRERMKAAGRRLQSFCVSLAQSFYQPLQRFFH  6177

Query  1948  DILHTKYRAATDVYALMFLAdvvdfiiiifgfwafgKHSAATDITSSLSDDQVPEAFLVM  2007
             DILHTKYRAATDVYALMFLAD+VD IIIIFGFWAFGKHSAATDI SSLSDDQVP+AFL M
Sbjct  6178  DILHTKYRAATDVYALMFLADIVDIIIIIFGFWAFGKHSAATDIASSLSDDQVPQAFLFM  6357

Query  2008  LLIQFSTMVVDRALYLRKTVLGKLAFQVALVLAIHLWMFFILPAVTERMFNQNVVAQLWY  2067
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Sbjct  6358  LLVQFGTMVIDRALYLRKTVLGKLAFQVVLVVAIHIWMFFILPAVTERMFSQNAVAQLWY  6537

Query  2068  FVKCIYFALSAYQIRCGYPTRILGNFLTKKYNHLNLFLFQGFRLVPFLVELRAVMDWVWT  2127
             FVKCIYFALSAYQIRCGYPTRILGNFLTKKYNHLNLFLFQGFRLVPFLVELRAVMDWVWT
Sbjct  6538  FVKCIYFALSAYQIRCGYPTRILGNFLTKKYNHLNLFLFQGFRLVPFLVELRAVMDWVWT  6717

Query  2128  DTTLSLSSWMCVEDIYANIFIIKCSRETEkkypqpkgqkkkkivkygMGGliilfliaii  2187
             DTTLSLS+WMCVEDIYANIFIIKCSRETEKKYPQPKGQKKKKIVKYGMGGLIILFLIAII
Sbjct  6718  DTTLSLSNWMCVEDIYANIFIIKCSRETEKKYPQPKGQKKKKIVKYGMGGLIILFLIAII  6897

Query  2188  wfpllfMSLVRSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQQPSIIPFTAQAYEELSRQF  2247
             WFPLLFMSL+RSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQQPSI+PFT QAYEELS+QF
Sbjct  6898  WFPLLFMSLIRSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQQPSIVPFTPQAYEELSQQF  7077

Query  2248  DPQPLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSRAQMKRELYNGTADITLRFTWNF  2307
             DP PLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSRAQMK+ELYNGTADITLRFTWNF
Sbjct  7078  DPYPLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSRAQMKQELYNGTADITLRFTWNF  7257

Query  2308  QRDLAKGGTVEYANEKHMLALAPNSTARRQLASLLEGTSDQSVVIPNLFPKYIRAPNGPE  2367
             QRDLAKGGTVEY NEKH L LAPNSTARRQLA LLEG  DQSVVIP+LFPKYIRAPNGPE
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Query  2368  ANPVKQLQPNEEADYLGVRIQLRREQ-GAGATG---------FLEWWVIELQECRTDCNL  2417
             ANPVKQLQP+EE DYLGVRIQLRREQ G GA+G         FLEWWVIELQ+C+ DCNL
Sbjct  7438  ANPVKQLQPDEEEDYLGVRIQLRREQVGTGASGEQAGTKASDFLEWWVIELQDCKADCNL  7617

Query  2418  LPMVIFSDKVSPPSLGFLAGYGIMGLYVSIVLVIGKFVRGFFSEISHSIMFEELPCVDRI  2477
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Query  2478  LKLCQDIFLVRETReleleeelYAKLIFLYRSPETMIKWTREKE  2521
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Sbjct  7798  LKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTRERE  7929
```

3. Chosen Protein:

```
>M. Musculus protein (sequence taken from BLAST result)
MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLLRALLCLSLLFLVA
HLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTRLVAPDLGVLLASSLCLGLCGRLTRKA
RQSRRTQELDDDDDDDDDDEDIDAAPAVGLKGAPALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGI
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LKNFVDLPNYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHELELDHLE
PEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLHGLGHLIMDQSYVCALIAMM
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MLYQLKIVNPHEYSSNCTEPFPNNTNLQPLEINQSLLYRGPVDPANWFGVRKGYPNLGYIQNHLQILLLLVFE
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ILHGCWLVAILTRRRREAIARLWPNYCLFLTLFLLYQYLLCLGMPPALCIDYPWRWSKAIPMNSALIKWLYLP
DFFRAPNSTNLISDFLLLLCASQQWQVFSAERTEEWQRMAGINTDHLEPLRGEPNPIPNFIHCRSYLDMLKVA
VFRYLFWLVLVVVFVAGATRISIFGLGYLLACFYLLLFGTTLLQKDTRAQLVLWDCLILYNVTVIISKNMLSL
LSCVFVEQMQSNFCWVIQLFSLVCTVKGYYDPKEMMTRDRDCLLPVEEAGIIWDSICFFFLLLQRRIFLSHYF
LHVSADLKATALQASRGFALYNAANLKSINFHRQIEEKSLAQLKRQMKRIRAKQEKYRQSQASRGQLQSK-DP
QDPSQEPGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEEEEALPEDPRPAAQSAFQMAYQAWVT
N---AQTVLRQRRERARQERAEQLASGGDLNPDVEPVDVPEDEMAGRSHMMQRVLSTMQFLWVLGQATVDGLT
RWLRAFTKHHRTMSDVLCAERYLLTQELLRVGEVRRGVLDQLYVGEDEATLSGPMETRDGPSTASSGLGAEEP
LSSMTDDTSSPLSTGYNTRSGSEEIVTDAGDLQAGTSLHGSQELLANARTRMRTASELLLDRRLHIPELEEAE
RFEAQQGRTLRLLRAGYQCVAAHSELLCYFIIILNHMVTASAASLVLPVLVFLWAMLTIPRPSKRFWMTAIVF
TEVMVVTKYLFQFGFFPWNSYVVLRRYENKPYFPPRILGLEKTDSYIKYDLVQLMALFFHRSQLLCYGLWDHE
EDRYPKDHCRSSVKDREAKEEPEAKLESQSETGTGHPKEPVLAGTPRDHIQGKGSIRSKDVIQDPPEDLKP-R
HTRHISIRFRRRKETPGPKGTAVMETEHEEGEGKETTERKRPRHTQEKSKFRERMKAAGRRLQSFCVSLAQSF
YQPLQRFFHDILHTKYRAATDVYALMFLADIVDIIIIIFGFWAFGKHSAATDIASSLSDDQVPQAFLFMLLVQ
FGTMVIDRALYLRKTVLGKLAFQVVLVVAIHIWMFFILPAVTERMFSQNAVAQLWYFVKCIYFALSAYQIRCG
YPTRILGNFLTKKYNHLNLFLFQGFRLVPFLVELRAVMDWVWTDTTLSLSNWMCVEDIYANIFIIKCSRETEK
KYPQPKGQKKKKIVKYGMGGLIILFLIAIIWFPLLFMSLIRSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQQP
SIVPFTPQAYEELSQQFDPYPLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSRAQMKQELYNGTADITLRF
TWNFQRDLAKGGTVEYTNEKHTLELAPNSTARRQLAQLLEGRPDQSVVIPHLFPKYIRAPNGPEANPVKQLQP
DEEEDYLGVRIQLRREQVGTGASGEQAGTKASDFLEWWVIELQDCKADCNLLPMVIFSDKVSPPSLGFLAGYG
IVGLYVSIVLVVGKFVRGFFSEISHSIMFEELPCVDRILKLCQDIFLVRETRELELEEELYAKLIFLYRSPET
MIKWTRERE
```
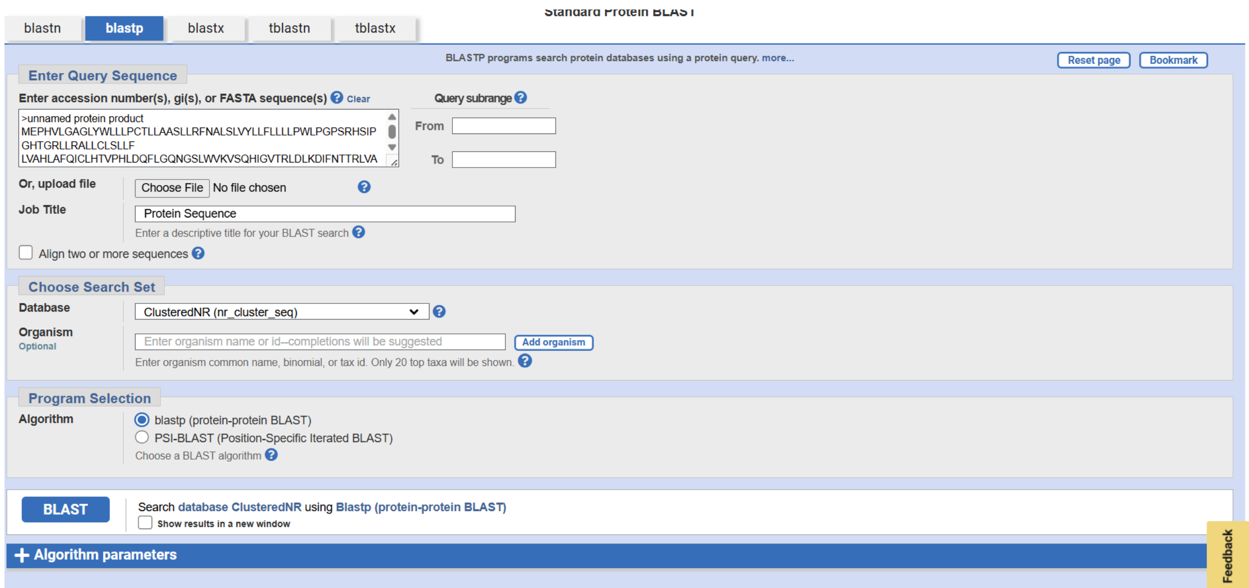
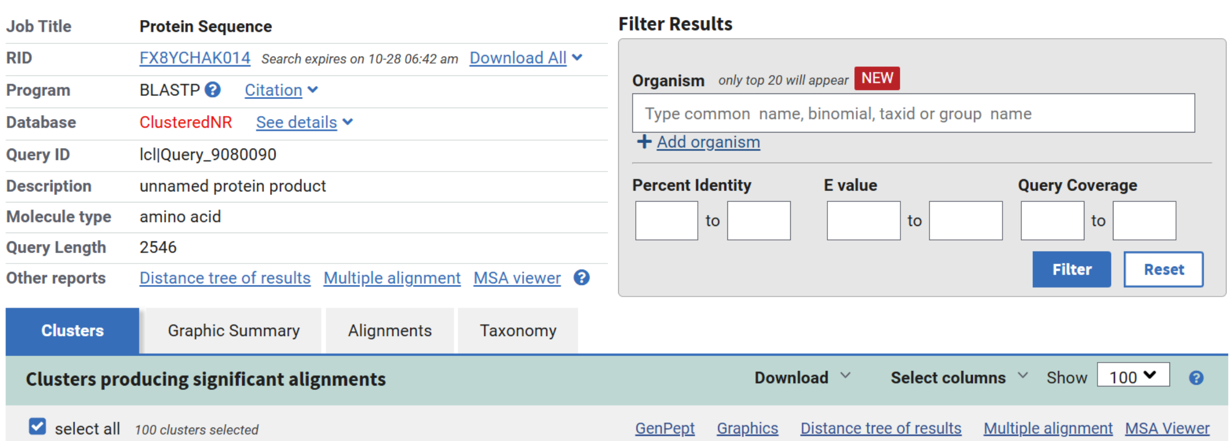

Name: piezo-type mechanosensitive ion channel component 1

Species: Mus musculus

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Glires; Rodentia; Myomorpha; Muroidea; Muridae; Murinae; Mus; Mus

4. A BLASTP search against NR database (see setup in first screen-shot below) yielded a top hit result to a protein from House Mouse. See additional screenshots below for top hits and selected alignment details:





Cluster Composition (Click the ⊞ to see the cluster contents)	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
⊞87 member(s), 29 organism(s)	house mouse	piezo-type mechanosensitive ion channel component 1 isofor...	5249	5249	100%	0.0	99.96%	2547	NP_001344278.1
⊞1 member(s), 1 organism(s)	large Japanese field mouse	Piezo-type mechanosensitive ion channel component [Apode...	4408	4408	100%	0.0	90.56%	2444	GAB1293756.1
⊞1 member(s), 1 organism(s)	Alston's brown mouse	hypothetical protein STEG23_002590 [Scotinomys teguina]	4401	4401	100%	0.0	90.20%	2514	KAL6082896.1
⊞1 member(s), 1 organism(s)	Bank vole	hypothetical protein U0070_027524 [Myodes glareolus]	4335	4335	97%	0.0	89.86%	2816	KAK7801803.1
⊞2 member(s), 2 organism(s)	southern multimammate ...	piezo-type mechanosensitive ion channel component 1 isofor...	4239	4239	84%	0.0	95.54%	2278	XP_031197536.1
⊞2 member(s), 1 organism(s)	lesser Egyptian jerboa	piezo-type mechanosensitive ion channel component 1 isofor...	4217	4217	100%	0.0	87.08%	2530	XP_045017799.1
⊞4 member(s), 1 organism(s)	American beaver	piezo-type mechanosensitive ion channel component 1 isofor...	4214	4214	100%	0.0	86.16%	2519	XP_073911579.1
⊞3 member(s), 2 organism(s)	woodchuck	Hypothetical predicted protein [Marmota monax]	4151	4151	100%	0.0	82.46%	2559	VTJ69060.1
⊞2 member(s), 2 organism(s)	woodchuck	piezo-type mechanosensitive ion channel component 1 isofor...	4144	4144	100%	0.0	82.82%	2502	XP_046291215.1

The top hit is from house mouse and has an E-value of 0.0, a 100% coverage, and

99.96% identity, which fulfills the criteria of a novel gene since the percent identity is less than 100%.

The alignment detail:

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piezo-type mechanosensitive ion channel component 1 isoform 1 [Mus musculus]

Sequence ID: NP_001344278.1 Length: 2547 Number of Matches: 1

Range 1: 1 to 2547 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps
5249 bits(13616)	0.0	Compositional matrix adjust.	2546/2547(99%)	2546/2547(99%)	1/2547(0%)

```
Query  1    MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL  60
            MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL
Sbjct  1    MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL  60

Query  61   RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  120
            RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR
Sbjct  61   RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  120

Query  121  LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQEL-DDDDDDDDDDEDIDAAPAVGLKGA  179
            LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQEL DDDDDDDDDDEDIDAAPAVGLKGA
Sbjct  121  LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQELQDDDDDDDDDDEDIDAAPAVGLKGA  180

Query  180  PALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTW  239
            PALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTW
Sbjct  181  PALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTW  240

Query  240  WSCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDL  299
            WSCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDL
Sbjct  241  WSCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDL  300

Query  300  PNYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHEL  359
            PNYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHEL
Sbjct  301  PNYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHEL  360

Query  360  ELDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLH  419
            ELDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLH
Sbjct  361  ELDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLH  420

Query  420  GLGHLIMDQSYVCALIAMMVWSIMYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLY  479
            GLGHLIMDQSYVCALIAMMVWSIMYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLY
Sbjct  421  GLGHLIMDQSYVCALIAMMVWSIMYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLY  480

Query  480  GLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFV  539
            GLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFV
Sbjct  481  GLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFV  540

Query  540  KEKLLKKQKVPAALLEVTVADTEPTQTQTLLRSLGELVTGIYVKYWIYVCAGMFIVVSFA  599
            KEKLLKKQKVPAALLEVTVADTEPTQTQTLLRSLGELVTGIYVKYWIYVCAGMFIVVSFA
Sbjct  541  KEKLLKKQKVPAALLEVTVADTEPTQTQTLLRSLGELVTGIYVKYWIYVCAGMFIVVSFA  600
```

Related Information

Gene - associated gene details

AlphaFold Structure - 3D structure displays

Genome Data Viewer - aligned genomic context